

# A Time Scenes Simulation Using Multiple Regression of the Color Block's Time Images on the Discrete Cosine Transform Domain

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## Abstract

In this paper, an efficient novel strategy simulating time scenes of an outdoor scene is introduced. It is a hybrid approach combining the multiple regression in statistic, the discrete cosine transform, and a look-up table algorithm. First, several color-block images taken at different times of a day were used as the testing images to create the reference database of the time scene. We then establish best fitting functions of light intensity in each plane (red, green, blue) with respect to time individually. These best fitting functions were generated by taking the multiple regressions on those coefficients of the DCT blocks of the color-block images. Finally, the time scenes simulation algorithm (TSSA) is able to synthesize the simulation scene at any assigned time by using the original input image.

To explore the utility and demonstrate the efficiency of the proposed algorithm, simulations under various input images are conducted. The experiment results show that our proposed algorithm can precisely simulate the desired scenes at an assigned time.

**Keywords:** Discrete Cosine Transform, Multiple Regressions, Color-block, Simulation.

## 1 Introduction

Sunrise and Sunset are the most beautiful scenes of a day, but they appear only in a very short time interval and people are almost always too busy to wait for their coming. If one can construct the sunset (or sunrise) scene of a scene image taken at anytime, then people shall no longer regret of missing the beautiful moments of their life.

The most special difference between a noon scene and a dusk (or dawn) scene is the colors of the sun and the sky. The Earth's atmosphere contains both air molecules and tiny aerosols

(much tinier than the wavelength of the visible light-ray from the Sun), both of which scatter the visible light-ray from the Sun which grazes the Earth's atmosphere. The scattering from both air molecules and aerosols is predominantly the Rayleigh scattering [1]- [4]. The intensity  $I$  of light scattered by a single tiny particle from a beam of light is given by

$$I = I_0 \frac{(1 + \cos^2 \Phi)}{2R^2} \left( \frac{2\pi}{\lambda} \right)^4 \left( \frac{n^2 - 1}{n^2 + 2} \right)^2 \left( \frac{d}{2} \right)^6 \quad (1)$$

where  $I_0$  is the original intensity of entrance light,  $\Phi$  is the scattering angle between the direction of incidence and the direction of observation,  $R$  is the large distance from the particle to the observer,  $\lambda$  is the wavelength of entrance light,  $n$  is the refractive index of particle, and  $d$  is the diameter of the particle. The Rayleigh scattering coefficient for a group of scattering particles is the multiplication product of the concentration of particles  $N$  and the cross-section  $\sigma_s$ .

$$\sigma_s = \frac{2\pi^5}{3} \frac{d^6}{\lambda^4} \left( \frac{n^2 - 1}{n^2 + 2} \right)^2 \quad (2)$$

According to the Rayleigh scattering law of light, blue light is scattered more strongly than red light. When the Sun is low in the sky, it appears less bright and redder than normal because more blue light than red light is scattered out of the solar light-rays, leaving an excess of red light. On the other hand, when the sun high in the sky, sunlight goes through very little atmosphere, so little scattering takes place, the sky far away to the sun appears light blue and the sky close to the sun appears mostly white (the sun's color).

It is very difficult to obtain the concentration and the diameters of the air molecules and the tiny aerosols in the Earth's atmosphere, so it is impossible to estimate the colors of an outdoor scene precisely by using the Rayleigh scattering law of light. In 2002, Thompson derived a spatial post-processing

algorithm of night scene [5] based on the visual noise (VN) and the loss of acuity (LA). There are four strategies in Thompson algorithm: (A) color space transform (CST), (B) day to night conversion (DNC), (C) night filter (NF), and (D) Gaussian noise generator (GNG). It can generate more realistic night scene, but it neglects the time factor so it cannot clearly illustrate the differences between early night, middle night and midnight. Alexander Toet used the method introduced by Welsh et al to transfer a source image's color characteristics to a grayscale target image (night vision imagery)[6,7]. They search a color source image that resembles the target scenes first. Second, they transformed both the source image and the target image into a perceptually miscorrelated color space. Third, they searched for the best matching source pixel for each target pixel from the source image by using the first order statistics of the luminance distribution in a  $7 \times 7$  window around the source and target pixels. Finally, they assigned the matching pixel's chromaticity values of the source image to the target pixel. Although they effectively give single-band intensified night vision gray imagery a full color day-time image, but to find a suitable resembled color source image for the target image is not easy, and the final colors of the chromatic target image completely depends on the colors of the source image. They are not the true colors of the target image taken in day. Other papers about the scenes can be found in [8]-[19]

The multiple regression is a related statistical method, it is used to model the relationships between several independent variables and a dependent variable by fitting a  $n$  equation to observed data. Numerous experiments have shown that the multiple regression models can be used to make accurate predictions. Multiple regression procedures are very widely used in the social and natural sciences today[20]-[22]. It might be a good and suitable method for us to simulate the time scene.

The spatial pixels of an image are often correlative, so the spatial domain of an image is not suitable for taking multiple regression. The DCT has properties of noise filtering, miscorrelation, and feature preservation. It is often used as a preprocessing step followed by a sophisticated algorithm in image processions, and it always has good performance while one combines the DCT block transformation with other techniques. In some cases, it has been showing that the processing based on DCT has significantly better results than the directive pixel based processing [24]. To take time scene simulation on the DCT domain by using multiple

regression will be a good idea. The remainder of this paper is organized as follows: Section 2 presents the the time scenes simulation algorithm (TSSA). Empirical results is described in the Section 3. Finally, Section 4 concludes this paper.

## 2 The Time Scenes Simulation

### Algorithm (TSSA)

#### 2.1 The construction of the multiple regression functions database

When performing experiment, we frequently tabulate data in the form of ordered pairs  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  with each  $x_i$  distinct. Given the data, it is then usually desirable to be able to predict  $y$  from  $x$  by finding a mathematical model, that is, a function  $y = f(x)$  that fits the data as closely as possible. One way to determine how well the function  $y = f(x)$  fits these order pairs  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  is to measure the sum of squares of the errors (SSE) between the predicted values of  $y$  and the observed values  $y_i$  for all of the  $n$  data points. Multiple linear regression is one of the widely used statistical techniques[20]-[22]. This technique is used to find a polynomial function of degree  $k$ ,  $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_k x^k$  as the predicting function, that has the minimum of the sum of squares of the errors(SSE) between the predicted values of  $y$  and the observed values  $y_i$  for all of the  $n$  data points  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ .

This paper proposed a way to construct the sunset scene of a scene image taken at anytime by TSSA. Figure.1 is the flow chart of the construction of the multiple regression functions database. At the beginning of the process, the R, G, and B planes of the decomposed color input image were split into non- overlapping  $8 \times 8$  blocks, and then transferred into the frequency domain from the spatial domain by block DCT respectively. In the DCT domain, most of the signal information tends to be concentrated in a few low-frequency components. And the DC coefficient carry information about average luminance within the DCT blocks, the DC coefficient becomes the most meaningful when

considering how the luminance is changing between the images[23,24]. We took the average of the DC components of all the  $8 \times 8$  DCT blocks to serve as the average value of the reflective color at that time. Then, we used multiple regression to find multiple regression functions of the DCT coefficients of these color-block images that were regularly photographed by fixed time interval from 17:00 to 19:30 of a day. These multiple regression functions were then constructed into a table as the database. Figure.2 shows the image of a color-block taken at 17:20, its R, G, B component images, and the images of DCT coefficients of the R, G, B component images.

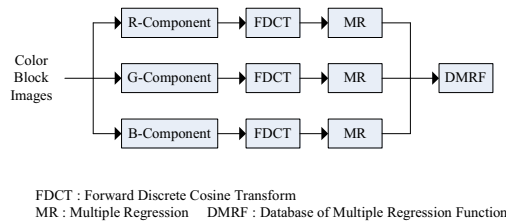


Figure 1. The construction of the multiple regression functions database

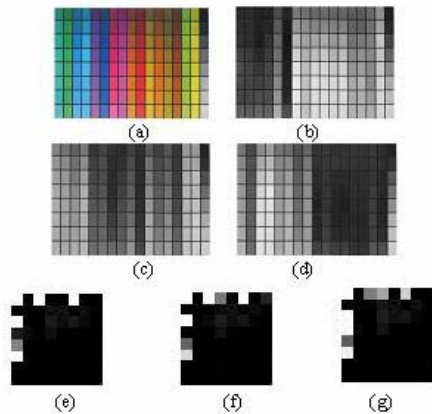


Figure 2. The image of a color-block taken at 17:20(a) real image (b) R component image(c) G component image (d) B component image (e)The image of DCT coefficients of the R component(f)The image of DCT coefficients of the G component(g)The image of DCT coefficients of the B component.

Table 1-3 are the tables of the average magnitudes of the  $8 \times 8$  DCT coefficients for the R, G, B component images of the color-block. There are 64 bases for each table, the upper left base is the block's mean and it is also called the DC value of the image, another 63 bases are called the AC values of the image. Table 1-3 show that the DC value is much larger than the AC values. And the higher frequencies AC values are lesser. Figure.3, Figure.4, and Figure.5 show parts of the curves of the best fitting functions for color-block's DC components with respect to time. Table 4 is the table of the best fitting functions' coefficients of the color-block image in DCT domain.

**Table 1 The DCT coefficients for the R component of the color-block image in Figure.2.**

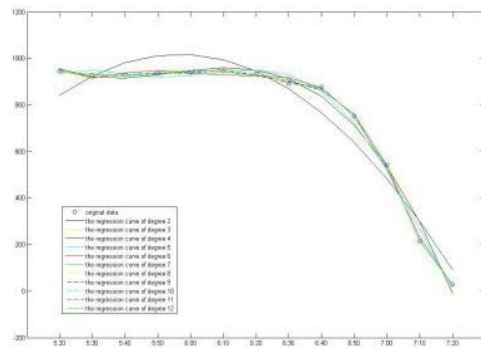
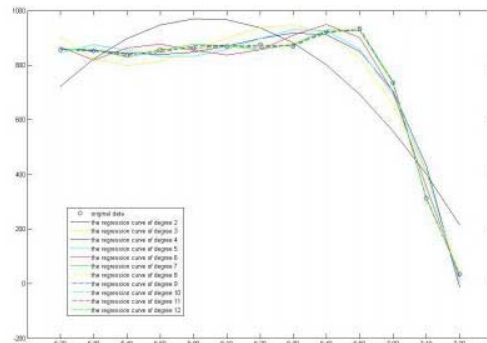
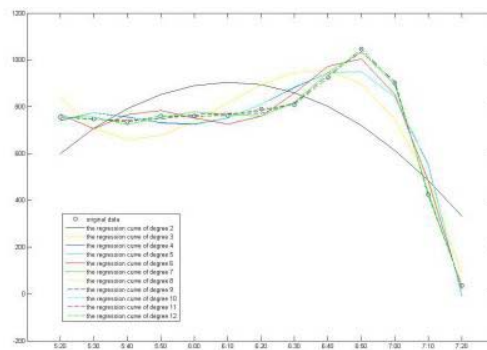
|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| 945.6788 | -3.11356 | 2.458706 | -0.33716 | -1.17563 | 1.24538  | -0.23217 | -0.927   |
| -5.97103 | -0.1986  | -0.02393 | 0.069363 | 0.06657  | -0.00409 | 0.057666 | -0.00114 |
| 16.23694 | -0.16898 | -0.09017 | 0.10713  | 0.059559 | 0.099738 | 0.041851 | -0.04458 |
| 0.121902 | 0.048323 | -0.05908 | 0.059259 | -0.03403 | 0.005014 | 0.003166 | -0.01487 |
| 0.504839 | -0.04899 | 0.000112 | -0.01591 | -0.01855 | 0.003226 | -0.00636 | 0.00413  |
| 1.000081 | -0.02425 | 0.00065  | -0.01323 | 0.0005   | -0.00932 | 0.000607 | 0.003872 |
| -0.23579 | -0.00222 | -0.02985 | -0.00037 | 0.003775 | 0.001865 | 0.000458 | 0.003585 |
| -0.48485 | -0.01236 | -0.00196 | 0.004875 | 0.008721 | -0.0107  | -0.00418 | 0.006822 |

**Table 2 The DCT coefficients for the G component of the color-block image in Figure.2.**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| 857.0752 | -1.16039 | 2.932919 | 0.480016 | -1.19657 | 1.780491 | -0.24423 | 0.147122 |
| -5.24403 | -0.09254 | -0.00674 | 0.0505   | 0.059852 | -0.00823 | 0.049707 | -0.01809 |
| 13.04633 | -0.1286  | 0.039403 | 0.10974  | 0.055904 | 0.090136 | 0.021017 | -0.02801 |
| -0.13035 | 0.011812 | -0.05759 | 0.071962 | -0.03963 | 0.006384 | 0.003408 | -0.01158 |
| 0.38559  | -0.03216 | 0.005886 | -0.01135 | -0.01836 | 0.004468 | 0.000146 | 0.00476  |
| 0.849799 | -0.02725 | -0.00554 | -0.01184 | -0.00021 | -0.01611 | -0.00312 | -0.00392 |
| -0.27928 | -0.01216 | -0.03458 | 0.009935 | 0.002494 | -0.01045 | 0.006994 | -0.00207 |
| -0.51225 | -0.00888 | -0.00602 | -0.0013  | 0.007954 | -0.00337 | 0.001674 | 0.006048 |

**Table 3 The DCT coefficients for the B component of the color-block image in Figure2.**

|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| 756.0942 | -1.61977 | 0.49209  | 0.649581 | -1.17632 | 0.806817 | -0.05249 | -1.09367 |
| -9.92855 | -0.18361 | -0.04784 | 0.088409 | 0.064015 | 0.004417 | 0.064835 | -0.0194  |
| 7.803619 | -0.11323 | -0.01613 | 0.111271 | 0.064356 | 0.089781 | 0.027291 | -0.05435 |
| 2.570166 | 0.024903 | -0.05725 | 0.078386 | -0.03275 | -0.00292 | 0.0057   | -0.01229 |
| 0.369813 | -0.0401  | -0.00135 | 0.000356 | -0.01725 | 0.002273 | -0.00308 | 0.003574 |
| 0.945137 | -0.02902 | -0.00551 | -0.00782 | 0.000382 | -0.0164  | 0.000275 | -6.6E-05 |
| -0.23514 | -0.01996 | -0.03039 | 0.009302 | 0.005829 | -0.00298 | 0.00189  | 0.010154 |
| -0.48137 | -0.01432 | -0.00211 | 0.005969 | 0.005762 | -0.00335 | -0.00572 | 0.005044 |

**Figure 3. Multiple regression curves of red component: the blue circle curve denotes the original data.****Figure 4. Multiple regression curves of green component: the blue circle curve denotes the original data.****Figure 5. Multiple regression curves of blue component: the blue circle curve denotes the original data.****Table 4 The table of the best fitting functions' coefficients of the color-block image's in DCT domain(a) R component (b) G component (c) B component.**

| (a)  |              |              |              |           |           |           |           |           |           |           |           |           |           |
|------|--------------|--------------|--------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|      | $\beta_{12}$ | $\beta_{11}$ | $\beta_{10}$ | $\beta_9$ | $\beta_8$ | $\beta_7$ | $\beta_6$ | $\beta_5$ | $\beta_4$ | $\beta_3$ | $\beta_2$ | $\beta_1$ | $\beta_0$ |
| 2th  |              |              |              |           |           |           |           |           |           |           | -13.296   | 123.774   | 730.627   |
| 3th  |              |              |              |           |           |           |           |           |           | -1.6349   | 21.0364   | -75.682   | 1005.29   |
| 4th  |              |              |              |           |           |           |           |           | 0.03917   | -2.7315   | 31.1697   | -110.07   | 1037.25   |
| 5th  |              |              |              |           |           |           |           | 0.05363   | -1.8379   | 21.134    | -102.10   | 199.743   | 818.438   |
| 6th  |              |              |              |           |           |           | 0.0193    | -0.7574   | 11.311    | -82.091   | 299.947   | -505.8    | 1226.7    |
| 7th  |              |              |              |           |           | 0.0037    | -0.1619   | 2.81641   | -24.995   | 119.47    | -297.114  | 339.238   | 805.771   |
| 8th  |              |              |              |           | 0.0003    | -0.01466  | 0.2649    | -2.5153   | 13.7158   | -45.305   | 96.805    | -132.12   | 1014.71   |
| 9th  |              |              |              | -8.3E-05  | 0.0055    | -0.154    | 2.3342    | -21.146   | 118.165   | -405.52   | 822.469   | -891.75   | 1321.28   |
| 10th |              |              | 3.9E-05      | -0.003    | 0.0884    | -1.5819   | 17.716    | -128.72   | 608.866   | -1832.9   | 3325.16   | -3241.7   | 2198.85   |
| 11th |              | 8.2E-05      | -0.0062      | 0.211     | -4.1037   | 51.0748   | -425.11   | 2401.3    | -9146.7   | 22841.6   | -35256.6  | 29861.1   | -9377.1   |
| 12th | 4.3E-05      | -0.00351     | 0.12828      | -2.7501   | 38.437    | -367.868  | 2463.6    | -11584.4  | 37773.8   | -82779.6  | 114789.6  | -89424.7  | 30039.4   |

| (b)  |              |              |              |           |           |           |           |           |           |           |           |           |           |
|------|--------------|--------------|--------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|      | $\beta_{12}$ | $\beta_{11}$ | $\beta_{10}$ | $\beta_9$ | $\beta_8$ | $\beta_7$ | $\beta_6$ | $\beta_5$ | $\beta_4$ | $\beta_3$ | $\beta_2$ | $\beta_1$ | $\beta_0$ |
| 2th  |              |              |              |           |           |           |           |           |           |           | -13.016   | 139.942   | 595.4     |
| 3th  |              |              |              |           |           |           |           |           |           | -2.7602   | 44.948    | -196.8    | 1059.1    |
| 4th  |              |              |              |           |           |           |           |           | -0.274    | 4.9109    | -25.931   | 43.743    | 835.55    |
| 5th  |              |              |              |           |           |           |           | 0.03605   | -1.5356   | 20.9512   | -115.51   | 251.97    | 688.48    |
| 6th  |              |              |              |           |           |           | 0.03127   | -1.2775   | 19.7592   | -146.22   | 535.617   | -890.66   | 1349.7    |
| 7th  |              |              |              |           |           | 0.00929   | -0.4239   | 7.70057   | -71.447   | 360.142   | -964.3    | 1232.21   | 292.22    |
| 8th  |              |              |              |           | 0.0006    | -0.0251   | 0.3756    | -2.2887   | 1.0791    | 51.4205   | -226.27   | 349.101   | 683.68    |
| 9th  |              |              |              | -0.0001   | 0.0079    | -0.2192   | 3.25888   | -28.248   | 146.613   | -450.48   | 784.83    | -709.34   | 1110.8    |
| 10th |              |              | -0.0001      | 0.0094    | -0.2812   | 4.75918   | -50.369   | 346.808   | -1564.22  | 4526.16   | -7940.8   | 7483.9    | -1948.8   |
| 11th |              | 1.5E-05      | -0.001       | 0.0487    | -1.0507   | 14.425    | -131.65   | 811.23    | -3355     | 9055.48   | -15023.1  | 13560.3   | -4073.8   |
| 12th | 2.7E-05      | -0.0023      | 0.0835       | -1.8171   | 25.7539   | -249.55   | 1688.5    | -8001.2   | 26209.5   | -57496.3  | 79520.9   | -61601.5  | 20762.6   |

| (c)  |              |              |              |           |           |           |           |           |           |           |           |           |           |
|------|--------------|--------------|--------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|      | $\beta_{12}$ | $\beta_{11}$ | $\beta_{10}$ | $\beta_9$ | $\beta_8$ | $\beta_7$ | $\beta_6$ | $\beta_5$ | $\beta_4$ | $\beta_3$ | $\beta_2$ | $\beta_1$ | $\beta_0$ |
| 2th  |              |              |              |           |           |           |           |           |           |           | -11.879   | 144.106   | 466.475   |
| 3th  |              |              |              |           |           |           |           |           |           | -3.6623   | 65.029    | -302.693  | 1081.74   |
| 4th  |              |              |              |           |           |           |           |           | -0.5827   | 12.6544   | -85.7341  | 208.952   | 606.224   |
| 5th  |              |              |              |           |           |           |           | 0.00263   | -0.6748   | 13.8244   | -92.2675  | 224.141   | 595.497   |
| 6th  |              |              |              |           |           |           | 0.0382    | -1.6014   | 25.331    | -190.331  | 702.892   | -1171.26  | 1402.95   |
| 7th  |              |              |              |           |           | 0.01184   | -0.542    | 9.84318   | -90.933   | 455.151   | -1209.11  | 1534.84   | 54.9832   |
| 8th  |              |              |              |           | 0.0013    | -0.0593   | 1.1121    | -10.824   | 59.1185   | -183.575  | 317.823   | -292.26   | 864.881   |
| 9th  |              |              |              | -9.6E-05  | 0.0074    | -0.2228   | 3.54005   | -32.684   | 181.671   | -606.224  | 1169.27   | -1183.6   | 1224.59   |
| 10th |              |              | -0.0002      | 0.01586   | -0.4765   | 8.1104    | -86.226   | 595.109   | -2682     | 7723.99   | -13436.3  | 12530.8   | -3896.9   |
| 11th |              | -7.8E-06     | 0.0004       | -0.0045   | -0.0781   | 3.10691   | -44.149   | 354.701   | -1755.1   | 5379.39   | -9770.2   | 9385.33   | -2796.9   |
| 12th | 3.5E-05      | -0.00294     | 0.1104       | -2.4258   | 34.7079   | -339.471  | 2317.98   | -11081.7  | 36612.8   | -80989.2  | 112925.6  | -88157.1  | 29434.8   |

## 2.2 The simulation image synthesis of time scene

Figure. 6 is the flowchart of the simulation image of time scene. To synthesize the scenes at specific time from the scene taken at any daytime, the input image was divided into R, G, and B planes firstly. The R, G, and B components of a scene photographed at time  $t_1$  are processed respectively with the same steps; they were transferred to the frequency domain by forward discrete cosine transform (FDCT) firstly. Then compared the DC component with the DC component of color-block at time  $t_1$  to

find the compensated coefficient  $C_1$ , to find the corresponding DC component  $f_{i,j}(t_2)$  at time  $t_2$  from looking up the database table, and then to combine  $f_{i,j}(t_2)$  and  $C_1$  to determine the final R, G, B components  $g_{i,j}(t_2)$  of the simulation scene at time  $t_2$  on frequency domain. Finally, we took the inverse discrete cosine transform (IDCT) on  $g_{i,j}(t_2)$ , and combined them to form the color simulation image at time  $t_2$  of the scene.

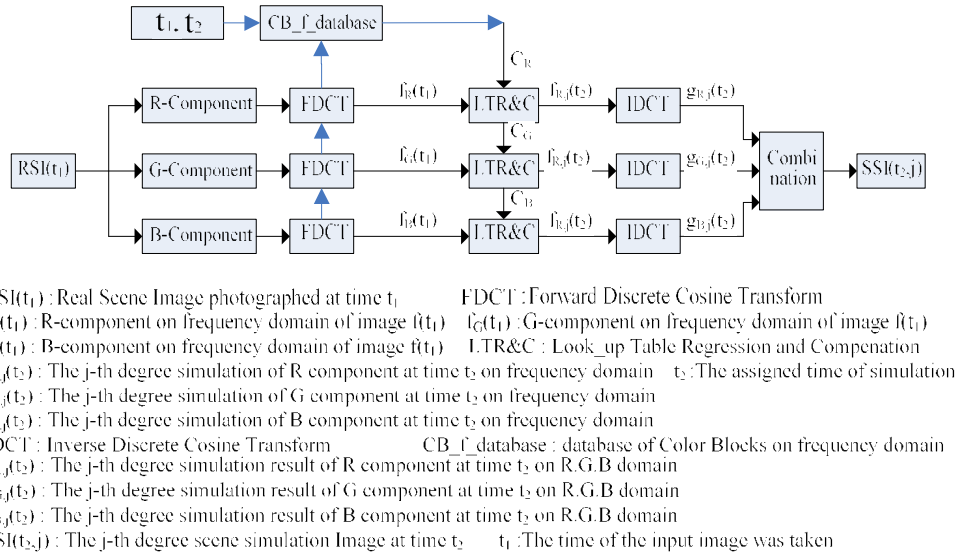


Figure 6. The flow chart of the simulation image synthesis of time scene.

### 3 Experiment results

Several test images with size  $(720 \times 512)$  were used in simulation for demonstrating the performance of the proposed scheme. In order to give a detailed description of the simulation results, Figure. 7 is used to express the simulation results; comparing the corresponding simulation image with the real image, we found some lamplights in the real image taken at 19:00, but not in the corresponding synthesis image. Apart from the lamplights, the human eyes can not feel any difference between them. Table 5 shows the absolute error ratios of simulation image with respect to the real image of Figure.7. From the error table we can find that the error in

R plane is the largest and the error in B plane is the smallest, this is caused by the fact that the average pixel value in B plane is larger than the average pixel value in R plane. On the other hand, the error in order 2 and order 3 best fitting regressions are much larger than the error in order larger than 3 best fitting regressions. This coincides with the Rayleigh scattering theory of sunlight. Experimental results show that there is no difference in visual quality between the simulation image obtained by our algorithm and the real image. However, because our algorithm is carried out in the DCT domain, it is time-saving and simpler.



Figure 7. Test images, (a) a scene photographed at 17:30 PM, (b)~(k) degree 2~degree 11 simulation scene at 19:00, (l) a scene photographed at 19:00 PM.

**Table 5 Table of absolute error ratios of simulation image with respect to real image of Figure.7.**

|                     | 2th    | 3th    | 4th    | 5th    | 6th    | 7th    | 8th    | 9th    | 10th     | 11th     | 12th     |
|---------------------|--------|--------|--------|--------|--------|--------|--------|--------|----------|----------|----------|
| Error in R plane(%) | 20.518 | 12.451 | 12.473 | 13.565 | 13.419 | 12.527 | 12.491 | 12.558 | 12.547   | 12.475   | 12.477   |
| Error in G plane(%) | 32.534 | 12.063 | 11.269 | 13.336 | 13.167 | 8.452  | 8.437  | 8.661  | 8.696    | 8.648    | 8.647    |
| Error in B plane(%) | 33.791 | 10.119 | 10.487 | 10.539 | 10.586 | 8.547  | 8.607  | 8.691  | 8.666    | 8.668    | 8.668    |
| Complexity          | $n^2$  | $n^3$  | $n^4$  | $n^5$  | $n^6$  | $n^7$  | $n^8$  | $n^9$  | $n^{10}$ | $n^{11}$ | $n^{12}$ |

#### 4 Conclusion

For any degree of best fitting simulation using a computer with Intel Pentium D 280GHz CPU and 512MB RAM, it only takes 3 seconds to simulate a scene image of size  $720 \times 512$  by our look-up table algorithm. On the other hand, simulating the same image by regression simulation with the same computer takes more

than 540 seconds. Table 1 is the table of absolute error ratios of simulation with respect to real image. It shows that the absolute error ratio of each degree simulations are always less than 16%; this error ratio can not be found by human eyes and it can be reduced by adopting more frequency bands in DCT domain of images for

regression and simulation. However, adopting more frequency bands is going to take more time and need more computer memory. With the above experimental results and performance analyses, we conclude that the proposed approach has three major advantages: (A) Using the color-block images instead of using the scene images of a place to create the reference database, this approach is suitable for any outdoor scene taken at anywhere at anytime. (B) Taking the multiple regression on the DCT coefficients of scene images instead of taking the multiple regression on the spatial pixels of scene images, this approach simplifies the regression procedure and saves the processing time. (C) Using look-up table algorithm to save time. With this algorithm, one can successfully and effectively simulate the anytime scene from a outdoor scene taken at anytime no matter it is of uniform color or complex colors, so it is very suitable to be used in the design of virtual reality and computer games.

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